

DKA & HHS

Hyperglycemic crises for IM residents — fluids, potassium, insulin, the gap

Step 1 — Diagnose & distinguish

2024 consensus criteria; watch for euglycemic DKA

	DKA	HHS
Glucose	≥200 (can be lower)	> 600
Ketones/acidosis	Present; bicarb cutoff 15	Minimal; pH > 7.3
Osmolality	Variable	> 320 effective
Mental status	Often alert	Often obtunded

PEARL Euglycemic DKA: SGLT2i / pregnancy / starvation — gap + ketones with near-normal glucose.

Step 2 — Fluids & the potassium gate

Balanced crystalloid first; check K before insulin

Serum K	Action
< 3.3	HOLD insulin — replace potassium first
3.3–5.2	Give insulin AND add potassium to fluids
> 5.2	Give insulin, recheck, hold potassium

PEARL Total-body K is depleted even when serum K looks normal — anticipate the fall once insulin starts.

Step 3 — Insulin & closing the gap

Treat the anion gap, not just the glucose

Principle	Detail
Target the gap	Resolve ketones/anion gap, not glucose alone
Add dextrose	At glucose ~200–250 to keep insulin running
Bicarbonate	Only if pH < 7.0 — not routine
Phosphate	Only if <1.0 with weakness/cardioresp compromise

PEARL Resolution = glucose <200 AND gap closed (bicarb $\geq$ 15–18, pH >7.3).

Find the trigger — the I's

- Infection
- Infarction (myocardial infarction)
- Insulin omission / nonadherence
- Ischemia
- Iatrogenic — steroids, SGLT2 inhibitors
- Also: new-onset diabetes, pregnancy

PEARL Work-up: BMP/gap, VBG, beta-hydroxybutyrate, osmolality, ECG/troponin, infection screen, HbA1c.

Transition — and what to avoid

DO

- Overlap SC basal insulin 1–2 h BEFORE stopping drip
- Use balanced crystalloid
- Respect the potassium gate
- Resolve the gap before declaring DKA over

AVOID

- Insulin when $K < 3.3$
- Routine bicarbonate (only $pH < 7.0$) / phosphate
- Stopping the drip with an open gap
- Being reassured by a normal glucose (euglycemic DKA)

PEARL The drip has no reservoir — an unbridged stop rebounds into ketoacidosis.

Take-home points

- Check ketones and the gap, not just glucose — euglycemic DKA is real
- Balanced fluids; potassium gate before insulin (hold if $K < 3.3$)
- Treat to gap closure; add dextrose at glucose ~ 200 – 250
- Bicarbonate only for $pH < 7.0$; phosphate only if severe + symptomatic
- Overlap basal insulin 1–2 h before stopping the drip; find the trigger